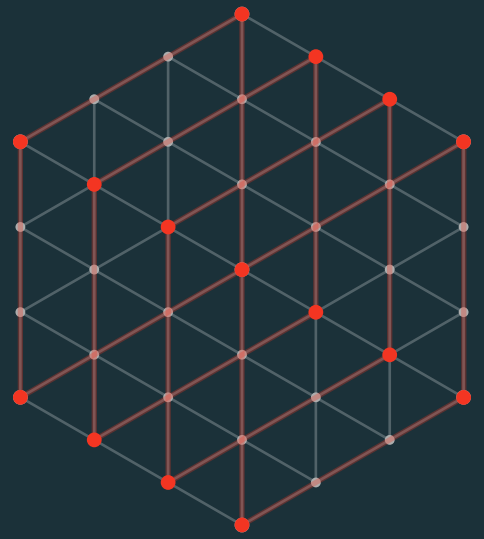
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ACME RETAIL



Real-time personalization on Databricks

The right product for every Acme Retail shopper within about 100 ms of a click, worth an expected $12.3M of revenue a year.

AGENDA

From the problem to a funded pilot

01 Acme Retail's challenge and who owns each outcome

02 The Databricks solution and architecture

03 Outcomes and the business case

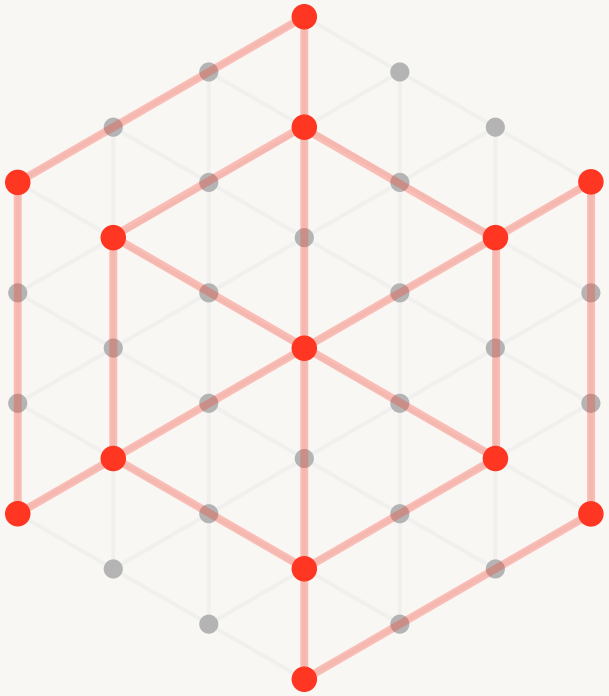
04 The pilot we propose

THE BUSINESS CHALLENGE

Acme Retail's paid traffic lands on a *generic* store

A first-time visitor and a loyal repeat buyer see the same homepage, rails, and offers.

- Low session-to-purchase conversion and high bounce on landing pages
- About 30% of sessions are cold start, with no history to personalize on
- Merchandisers can't see why a product was shown, or whether it worked
- Personalization logic lives in app code, away from the analytics the business trusts



WHO OWNS THE OUTCOME

Every metric sits on an Acme Retail leader's scorecard

01

VP of E-commerce, pilot sponsor

Owns session conversion, revenue per session and AOV: the online revenue growth target.

02

CFO, funding approver

Owns ROI, payback and run cost, and co-owns AOV and margin mix with E-commerce.

03

VP of Merchandising

Owns recommendation CTR and category performance, and needs answers without an analyst ticket.

04

CMO

Owns return on paid acquisition, so cold-start shoppers who buy on their first visit move this number.

WHY IT IS HARD TODAY

A stitched stack makes every step a hand-off

Need	Stitched stack	Databricks
Fresh signals	Nightly batch features	Last 5 clicks read from Lakebase per request
Serving data	Separate database, custom ETL	Gold tables synced to Lakebase automatically
Governance	Rules re-built per system	Unity Catalog masks, filters, lineage
Explainability	None for the business	Grounded "why this?" for every recommendation
Insight	Analyst ticket, hours	Genie and a copilot, seconds

SECTION 02

The Databricks solution, from click to recommendation



THE SOLUTION

Three capabilities on one governed platform

Real-time signal

Lakeflow streams clickstream into governed tables. Lakebase serves features and live session state in milliseconds.

Intelligence

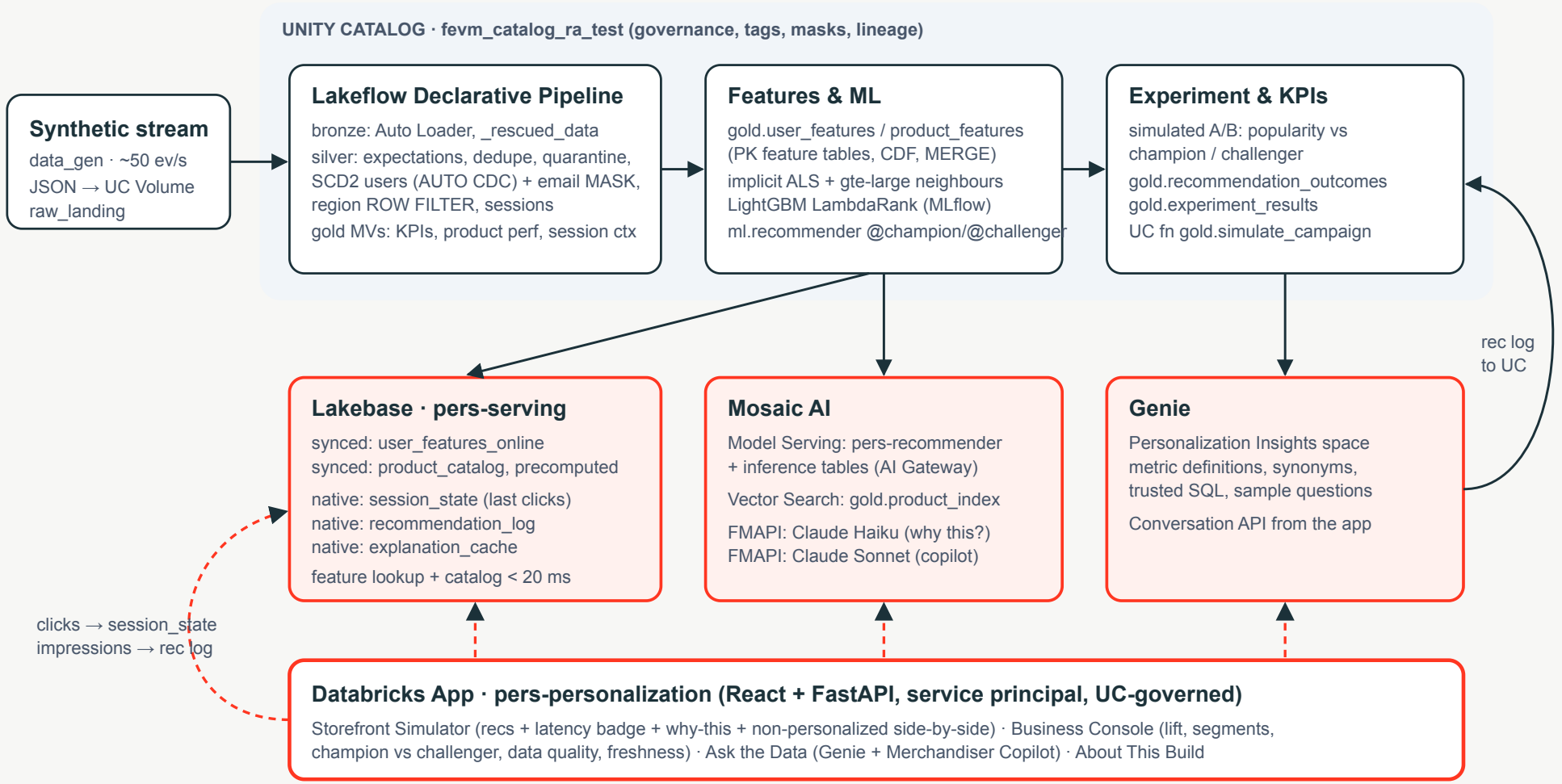
A two-stage recommender on Model Serving, Vector Search for cold start, and grounded LLM explanations.

Business loop

An A/B experiment, a business case in dollars, a Genie space, and a merchandiser copilot.

ARCHITECTURE

Every stage consumes the previous stage's output



HOW IT WORKS

Four steps turn a click into the next recommendation

1

Capture the click

The app writes each view, add-to-cart, and purchase to Lakebase session state.

2

Look up context

User features and the last 5 clicks come back from Lakebase in about 5 ms.

3

Rank candidates

ALS and embedding candidates are ranked by LightGBM on Model Serving.

4

Explain and log

Claude writes a grounded "why this?" and every impression is logged for attribution.

BUILT FOR PRODUCTION

Governed, resilient, and explainable by design

01

Governed end to end

Email masked and tagged as PII, regional row filters, and lineage from landing volume to served model in Unity Catalog.

02

Resilient serving

If Model Serving is slow, precomputed recommendations in Lakebase fill the rail, so shoppers never see an empty page.

03

Explainable and measured

Champion and challenger models in Unity Catalog, an A/B readout, and a copilot scored 20 of 20 on grounded answers.

HOW WE BUILT IT

An AI agent built every layer; the architect made every call

Layer	What the AI agent did	Human decision, and a fix caught in testing
Data and pipeline	Wrote the persona simulator, Lakeflow pipeline and masks	Set the 3% CTR baseline; first calibration gave 0.03% conversion, re-tuned in 3 runs
Models	Wrote ALS, embeddings and the LightGBM ranker in MLflow	Chose two-stage design; fixed an artifact crash from Spark Connect metadata
Serving	Wrote the endpoint, Lakebase sync and fallback path	Found every live call silently falling back; fixed the response parser
GenAI	Wrote the copilot, explanations and a 20-question eval	Grounding judge scored 10% without evidence; fixed to 20 of 20

Agent: Claude Code. In the product: Foundation Model APIs, Vector Search and Genie. Full part-by-part log in docs/decisions.md.

SECTION 03

Outcomes and the business case

PERSONALIZED VS POPULARITY BASELINE

Personalization doubles engagement

2x

recommendation CTR, 3.1% to 6.1%, for the VP of Merchandising

+47%

session conversion, 1.93% to 2.84%, for the VP of E-commerce

2x

cold-start conversion, 0.83% to 1.67%, for the CMO

Simulated A/B on synthetic data; outcomes drawn from hidden shopper preferences the model never sees

SCORECARD AGAINST TARGETS

Engagement targets are met; AOV and latency are the next focus

Metric	Owner	Baseline	Personalized	Target
Recommendation CTR	VP Merchandising	3.1%	**6.1%**	at least 6%
Session conversion	VP E-commerce	1.93%	**2.84% to 3.30%**	at least 2.6%
Cold-start conversion	CMO	0.83%	**1.67%**	at least 1.4%
Average order value	VP E-commerce, CFO	$57.92	**$56.47 to $60.34**	plus 5 to 8%
Latency	VP E-commerce	n/a	**about 112 ms p50**	under 100 ms p95

Returning shoppers unless noted. Range shows champion to challenger. Offline NDCG@10 rose from 0.072 to 0.121.

THE BUSINESS CASE, FOR THE CFO

An expected *$4.3M* of gross profit a year

ACME RETAIL ASSUMPTIONS

- Revenue per session rises $0.41, from $0.95 to $1.36 (measured)
- 5M sessions a month and 35% gross margin (replace with actuals)
- Only 50% of the simulated lift is banked
- $750k a year run cost for platform and team

ANNUAL VALUE, CHAMPION MODEL

- $12.3M incremental revenue and $4.3M gross profit
- $3.5M net benefit, a 5.7x return on run cost
- Payback in about 2 months
- Low case still pays: $2.1M gross profit, 4.2-month payback

Value = revenue-per-session lift x annual sessions x realization x margin, minus run cost. Computed by UC function gold.business_case.

THE PILOT WE PROPOSE

Two weeks on Acme Retail's own clickstream

- Live A/B on a slice of traffic, judged on each owner's own number
- Replaces the 50% realization assumption with a measured lift
- Swap the synthetic feed for Acme Retail's events; keep the rest

```
# deploy pipeline, jobs, Lakebase, app
databricks bundle deploy

# data, features, models, serving, Genie
databricks bundle run pers_setup
databricks bundle run pers_app
```

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Thank you

The ask: approve the two-week Acme Retail pilot, sponsored by the VP of E-commerce.

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